

B3.2

P1. Solution:

$$\left(\frac{5}{2}\right)(2x) + \left(\frac{8}{3}\right)(3x^2) + (-1)(5x - 8x^2) = 0$$

$$\frac{5}{2}f(x) - \frac{8}{3}g(x) - h(x) = 0$$

P4. Solution:

$$-\frac{3}{17}f(x) - \frac{2}{2}g(x) + h(x) = 0$$

P7. Solution:

$$W(f, g, h) = \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & 2x \\ 0 & 0 & 2 \end{vmatrix}$$

$$= 2 \neq 0$$

P8. Solution:

$$W(f, g, h) = \begin{vmatrix} e^x & e^{2x} & e^{3x} \\ e^x & 2e^{2x} & 3e^{3x} \\ e^x & 4e^{2x} & 9e^{3x} \end{vmatrix}$$

$$= 2e^{6x} \neq 0$$

P10. Solution:

$$W(f, g, h) = \begin{vmatrix} e^x & x^{-2} & x^{-2} \ln x \\ e^x & -2x^{-3} & -2x^{-3} \ln x + x^{-3} \\ e^x & 6x^{-4} & 6x^{-4} \ln x - 5x^{-4} \end{vmatrix}$$

$$= e^x (4x^{-7} + 5x^{-6} + x^{-5})$$

$$\neq 0 \text{ when } x > 0.$$

P13. Solution: $\rightarrow y_1 \quad \rightarrow y_2 \quad \rightarrow y_3$

$$y(x) = C_1 e^x + C_2 e^{-x} + C_3 e^{-2x}$$

$$y'(x) = C_1 e^x - C_2 e^{-x} - 2C_3 e^{-2x}$$

$$y''(x) = C_1 e^x + C_2 e^{-x} + 4C_3 e^{-2x}$$

$$\begin{aligned} y(0) &= C_1 + C_2 + C_3 = 1 \\ y'(0) &= C_1 - C_2 - 2C_3 = 2 \\ y''(0) &= C_1 + C_2 + 4C_3 = 0 \end{aligned} \Rightarrow \begin{aligned} C_1 &= \frac{4}{3} \\ C_2 &= 0 \\ C_3 &= -\frac{1}{3} \end{aligned}$$

$$\text{Thus, } y(x) = \frac{4}{3}e^x - \frac{1}{3}e^{-2x}$$

P17. Solution:

$$y(x) = C_1 + C_2 \cos 3x + C_3 \sin 3x$$

$$y'(x) = -3C_2 \sin 3x + 3C_3 \cos 3x$$

$$y''(x) = -9C_2 \cos 3x - 9C_3 \sin 3x$$

$$\begin{aligned} y(0) &= C_1 + C_2 = 3 \\ y'(0) &= 3C_3 = -1 \\ y''(0) &= -9C_2 = 2 \end{aligned} \Rightarrow \begin{aligned} C_1 &= \frac{29}{9} \\ C_2 &= -\frac{2}{9} \\ C_3 &= -\frac{1}{3} \end{aligned}$$

$$\text{Thus, } y(x) = \frac{29}{9} - \frac{2}{9} \cos 3x - \frac{1}{3} \sin 3x$$

P19. Solution:

$$y(x) = C_1 x + C_2 x^2 + C_3 x^3$$

$$y'(x) = C_1 + 2C_2 x + 3C_3 x^2$$

$$y''(x) = 2C_2 + 6C_3 x$$

$$\begin{aligned} y(1) &= C_1 + C_2 + C_3 = 6 \\ y'(1) &= C_1 + 2C_2 + 3C_3 = 14 \\ y''(1) &= 2C_2 + 6C_3 = 22 \end{aligned} \Rightarrow \begin{aligned} C_1 &= 1 \\ C_2 &= 2 \\ C_3 &= 3 \end{aligned}$$

$$\text{Thus } y(x) = x + 2x^2 + 3x^3$$

P21. Solution:

$$y(x) = y_c(x) + y_p(x)$$

$$= C_1 \cos x + C_2 \sin x + 3x$$

$$y'(x) = -C_1 \sin x + C_2 \cos x + 3$$

$$\begin{aligned} y(0) &= C_1 = 2 \\ y'(0) &= C_2 + 3 = -2 \end{aligned} \Rightarrow \begin{aligned} C_1 &= 2 \\ C_2 &= -5 \end{aligned}$$

$$\text{Thus, } y(x) = 2 \cos x - 5 \sin x + 3x$$

P24. Solution:

$$y(x) = y_c + y_p = C_1 e^x \cos x + C_2 e^x \sin x + x + 1$$

$$y'(x) = C_1 e^x \cos x - C_1 e^x \sin x + C_2 e^x \sin x + C_2 e^x \cos x + 1$$

$$= C_1 e^x (\cos x - \sin x) + C_2 e^x (\sin x + \cos x) + 1$$

$$\Rightarrow y(0) = C_1 + 1 = 4 \quad \Rightarrow C_1 = 3$$

$$y'(0) = C_1 + C_2 + 1 = 8 \quad \Rightarrow C_2 = 4$$

$$\text{Thus, } y(x) = 3e^x \cos x + 4e^x \sin x + x + 1.$$

S3.3.

P1. Solution: Characteristic equation:

$$r^2 - 4 = 0$$

$$\Rightarrow r_1 = 2, r_2 = -2$$

$$\Rightarrow y_1(x) = e^{2x}, y_2(x) = e^{-2x}$$

$$\text{Thus, } y(x) = C_1 e^{2x} + C_2 e^{-2x}$$

P2. Solution: Characteristic equation:

$$2r^2 - 3r = 0$$

$$r(2r - 3) = 0$$

$$r_1 = 0, r_2 = \frac{3}{2}$$

$$\Rightarrow y_1(x) = 1, y_2 = e^{\frac{3}{2}x}$$

$$\text{Thus, } y(x) = C_1 + C_2 e^{\frac{3}{2}x}$$

P4. Solution: Characteristic equation:

$$2r^2 - 7r + 3 = 0$$

$$(2r - 1)(r - 3) = 0$$

$$r_1 = \frac{1}{2}, r_2 = 3$$

$$\Rightarrow y_1(x) = e^{\frac{1}{2}x}, y_2(x) = e^{3x}$$

$$\text{Thus, } y(x) = C_1 e^{\frac{1}{2}x} + C_2 e^{3x}$$

P8. Solution: Characteristic equation:

$$r^2 - 6r + 13 = 0$$

$$r = 3 \pm 2j$$

$$\text{Thus, } y(x) = e^{3x} (C_1 \cos 2x + C_2 \sin 2x)$$

P11. Solution: Characteristic equation:

$$r^4 - 8r^3 + 16r^2 = 0$$

$$r^2(r^2 - 8r + 16) = 0$$

$$r^2(r - 4)^2 = 0$$

$$\Rightarrow \begin{cases} r_1 = 0 & (\text{double root } k=2) \\ r_2 = 4 & (\text{double root } k=2) \end{cases}$$

$$r_1 = 0 \text{ contributes } (C_1 + C_2 x) e^{0x} = (C_1 + C_2 x)$$

$$r_2 = 4 \text{ contributes } (C_3 + C_4 x) e^{4x}$$

$$\text{Thus, } y(x) = C_1 + C_2 x + (C_3 + C_4 x) e^{4x}$$

P14. Solution: Characteristic equation

$$r^4 + 3r^2 - 4 = 0$$

$$(r^2 + 4)(r^2 - 1) = 0$$

$$\Rightarrow r^2 = -4 \text{ or } r^2 = 1$$

$$\Rightarrow r_{1,2} = \pm 2i, r_3 = 1, r_4 = -1.$$

$$r_{1,2} = \pm 2i \text{ contribute } C_1 \cos 2x + C_2 \sin 2x$$

$$r_3 = 1 \text{ contributes } C_3 e^x$$

$$r_4 = -1 \text{ contributes } C_4 e^{-x}$$

$$\text{Thus, } y(x) = C_1 \cos 2x + C_2 \sin 2x + C_3 e^x + C_4 e^{-x}$$

P16. Solution: Characteristic equation.

$$r^4 - 8r^2 + 16 = 0$$

$$(r^2 - 4)^2 = 0$$

$$(r - 2)^2 (r + 2)^2 = 0$$

$$r_1 = 2 \text{ (double root)}$$

$$r_2 = -2 \text{ (double root)}$$

$$r_1 = 2 \text{ Contributes } (C_1 + C_2 x)e^{2x}$$

$$r_2 = -2 \text{ Contributes } (C_3 + C_4 x)e^{-2x}$$

$$\text{Thus, } y(x) = (C_1 + C_2 x)e^{2x} + (C_3 + C_4 x)e^{-2x}$$

P21. Solution: Characteristic equation:

$$r^2 - 4r + 3 = 0$$

$$\Rightarrow (r-3)(r-1) = 0$$

$$\Rightarrow r_1 = 3, r_2 = 1$$

$$\Rightarrow y(x) = C_1 e^{3x} + C_2 e^x$$

$$y'(x) = 3C_1 e^{3x} + C_2 e^x$$

$$y(0) = C_1 + C_2 = 7 \quad \left. \begin{array}{l} \\ \end{array} \right\} \Rightarrow \begin{array}{l} C_1 = 2 \\ C_2 = 5 \end{array}$$

$$y'(0) = 3C_1 + C_2 = 11 \quad \left. \begin{array}{l} \\ \end{array} \right\}$$

$$\text{Thus, } y(x) = 2e^{3x} + 5e^x$$

P24. Solution: Characteristic equation:

$$2r^3 - 3r^2 - 2r = 0$$

$$r(2r^2 - 3r - 2) = 0$$

$$r(2r+1)(r-2) = 0$$

$$\Rightarrow r_1 = 0, r_2 = -\frac{1}{2}, r_3 = 2$$

$$\text{Thus, } y(x) = C_1 + C_2 e^{-\frac{1}{2}x} + C_3 e^{2x}$$

$$y'(x) = -\frac{1}{2}C_2 e^{-\frac{1}{2}x} + 2C_3 e^{2x}$$

$$y''(x) = \frac{1}{4}C_2 e^{-\frac{1}{2}x} + 4C_3 e^{2x}$$

$$y(0) = C_1 + C_2 + C_3 = 1 \quad \left. \begin{array}{l} \\ \end{array} \right\} \Rightarrow \begin{array}{l} C_1 = -\frac{7}{2} \\ C_2 = 4 \\ C_3 = \frac{1}{2} \end{array}$$

$$y'(0) = -\frac{1}{2}C_2 + 2C_3 = -1 \quad \left. \begin{array}{l} \\ \end{array} \right\}$$

$$y''(0) = \frac{1}{4}C_2 + 4C_3 = 3 \quad \left. \begin{array}{l} \\ \end{array} \right\}$$

$$\text{Thus, } y(x) = -\frac{7}{2} + 4e^{-\frac{1}{2}x} + \frac{1}{2}e^{2x}$$

P26. Solution: Characteristic equation:

$$r^3 + 10r^2 + 25r = 0$$

$$r(r+5)^2 = 0$$

$$r = 0, r = -5 \text{ (double root)}$$

$$\text{Thus } y(x) = C_1 + (C_2 + C_3 x)e^{-5x}$$

$$y'(x) = -5C_2 e^{-5x} + C_3 [1 - 5x]e^{-5x}$$

$$y''(x) = 25C_2 e^{-5x} + C_3 [-10 + 25x]e^{-5x}$$

$$\Rightarrow y(0) = C_1 + C_2 = 3 \quad \left. \begin{array}{l} \\ \end{array} \right\} \Rightarrow \begin{array}{l} C_1 = \frac{24}{5} \\ C_2 = -\frac{9}{5} \end{array}$$

$$y'(0) = -5C_2 + C_3 = 4 \quad \left. \begin{array}{l} \\ \end{array} \right\}$$

$$y''(0) = 25C_2 - 10C_3 = 5 \quad C_3 = -5$$

$$\Rightarrow y(x) = \frac{24}{5} + (-\frac{9}{5} - 5x)e^{-5x}$$

P33. Solution: Characteristic equation:

$$r^3 + 3r^2 - 54 = 0$$

$y = e^{3x}$ is a solution $\Rightarrow r = 3$ is a root of the characteristic equation.

$$(r-3)(r^2 + 6r + 18) = 0$$

$$\Rightarrow r_1 = 3, r_{2,3} = -3 \pm 3i$$

$$r_{2,3} = -3 \pm 3i \text{ Contribute } e^{-3x}(C_2 \cos 3x + C_3 \sin 3x)$$

$$\text{Thus, } y(x) = C_1 e^{3x} + e^{-3x}(C_2 \cos 3x + C_3 \sin 3x)$$